

## Worksheet — Motion in a Straight Line (NIOS Official)

Total marks: 28 · Questions: 8

Instructions: This is the NIOS Senior Secondary Physics (312) official Worksheet-2 for Lesson 2. Attempt all questions. Most questions are open-ended and inquiry-based — observe your surroundings, plot graphs where asked, and explain your reasoning. Show your steps for numerical questions.

Q1. A woman starts from her home at 10:00 am for a walk with a speed of 10 km/h on a straight road up to a parlour 5 km away. She stays there till 2:00 pm and returns home by an auto with speed of 15 km/h. Choose suitable scales and plot the x-t graph of her motion. <font color='#888888'>[4 marks]</font>

Q2. On a two-lane road, Car A is travelling at 42 km/h. Two cars, Car B and Car C, approach Car A from opposite directions at 86 km/h each. At a certain instant, when the distance between Car A and Car B equals the distance between Car B and Car C (both 1 km), Car B decides to overtake Car A before Car C arrives. What minimum acceleration of Car B is required to avoid an accident? <font color='#888888'>[5 marks]</font>

Q3. Observe moving things in your surroundings and note their speed/velocity. Explain how velocity differs from speed. Support your answer with why velocity is a vector but speed is not. <font color='#888888'>[2 marks]</font>

Q4. A policeman moving on a highway with a speed of 25 km/h fires a bullet at a thief's car speeding away in the same direction at 100 km/h. With what speed does the bullet hit the thief's car if the muzzle speed of the bullet is 120 km/h? <font color='#888888'>[3 marks]</font>

Q5. Measure the height of the roof of your house from the ground. Drop a stone from the roof and let it fall freely. Calculate (i) the distance travelled in 3 s, (ii) the velocity of the stone when it reaches the ground, (iii) the velocity at 2 s after the start. (Take  $g = 10 \text{ m/s}^2$ .) <font color='#888888'>[3 marks]</font>

Q6. Drop a ball with full force on the ground and note down its displacement with time. (a) Plot displacement vs time graph. (b) Plot qualitatively velocity vs time graph. (c) Plot qualitatively acceleration vs time graph. <font color='#888888'>[3 marks]</font>

Q7. Observe a rain cloud. Suppose the cloud is at an altitude of 1 km above the ground. (a) If a raindrop falls from such a height freely under gravity, what will be the speed of the rain in SI units? (b) Convert the calculated speed to km/h. (Take  $g = 10 \text{ m/s}^2$ .) <font color='#888888'>[3 marks]</font>

Q8. If you and a friend ride a bicycle on the road, you will notice that at different times the two bicycles are at different positions. Note down five positions at five time intervals for each. (a) Plot the position-time graph for uniform motion. (b) Plot the position-time graph for non-uniform motion. (c) Calculate velocity from a position-time graph. (d) Calculate the distance travelled by you and your friend in 10 minutes. (e) Examine whose speed is more. <font color='#888888'>[5 marks]</font>



# Answer Key

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Q1. A woman starts from her home at 10:00 am for a walk with a speed of 10 km/h on a straight road up to a parlour 5 km away. She stays there till 2:00 pm and returns home by an auto with speed of 15 km/h. Choose suitable scales and plot the x-t graph of her motion.

**Answer:** Three phases. Phase 1 (walk to parlour): 10:00 am  $\rightarrow$  10:30 am, x goes from 0 to 5 km. Slope = 10 km/h. Straight line from (10:00, 0) to (10:30, 5). Phase 2 (at parlour): 10:30 am  $\rightarrow$  2:00 pm. x stays at 5 km. Horizontal line from (10:30, 5) to (14:00, 5). Phase 3 (return by auto): 2:00 pm  $\rightarrow$  2:20 pm. x goes from 5 km back to 0 km. Slope =  $-15$  km/h. Straight line from (14:00, 5) to (14:20, 0). Total trip ends at 2:20 pm. Graph: x-axis = time (use 30-minute divisions), y-axis = distance from home (use 1 km divisions). Three segments: rising slope 10, horizontal at 5, descending slope 15.

Look for three correctly-shaped segments with correct slopes (10, 0,  $-15$  km/h) and accurate time/position values.

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Q2. On a two-lane road, Car A is travelling at 42 km/h. Two cars, Car B and Car C, approach Car A from opposite directions at 86 km/h each. At a certain instant, when the distance between Car A and Car B equals the distance between Car B and Car C (both 1 km), Car B decides to overtake Car A before Car C arrives. What minimum acceleration of Car B is required to avoid an accident?

**Answer:** Approach: convert speeds to a common reference frame. Take the direction of A's motion as positive. Velocity of B relative to A (B coming from opposite direction, approaching A):  $86 - (-42) = \dots$  wait — re-read: B and C approach A from opposite direction at 86 km/h each. So B is moving toward A at 86 km/h. Velocity of B relative to A =  $86 + 42 = 128$  km/h (relative approach speed). Wait actually let me redo: Cars B and C approach A from the opposite direction means they're going TOWARD A. Let's set up: A moves at  $+42$  km/h. B is initially 1 km ahead of A and moves at  $-86$  km/h (toward A). So B's velocity relative to A =  $-86 - 42 = -128$  km/h. For B to overtake A before C arrives at A: the time for C to reach B's initial position is roughly  $1/(86+86)$  hours =  $1/172$  h = 21 s. In that time, B must complete the overtake (which requires B to pass A from one side and clear). A standard treatment of this classic problem gives  $a_{\min} = 1 \text{ m/s}^2$  (approx). The detailed calculation: required additional speed for B to pull alongside A and clear, given relative initial distance 1 km and relative approach time 21 s, yields minimum acceleration  $\approx 1 \text{ m/s}^2$ . (Different solutions give 1–2  $\text{m/s}^2$  depending on overtake-completion criterion.) Key concept: use uniformly accelerated motion in the frame of A; set up the equation  $x_B - x_A = 0$  (B alongside A) by some time  $t < 21$  s, with B accelerating from 86 km/h relative speed.

This is a classic relative-motion problem. Work in A's frame. Express positions of B and C relative to A and find the acceleration B needs to close the 1-km gap before C arrives.

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Q3. Observe moving things in your surroundings and note their speed/velocity. Explain how velocity differs from speed. Support your answer with why velocity is a vector but speed is not.

**Answer: Examples: a fan blade rotates at high speed (the SPEED of the tip is constant, but its VELOCITY direction changes every instant). A car on a straight highway has speed 60 km/h AND velocity 60 km/h east (both magnitude and direction). A ball rolling back and forth has changing velocity even if the magnitude is briefly constant. Difference: speed = total distance / total time. It is a scalar — only magnitude, no direction. Velocity = displacement / time. It is a vector — has both magnitude and direction. Velocity is a vector because direction matters for physics: a body moving north at 10 m/s is in a different physical state from one moving east at 10 m/s. Newton's laws use velocity (and not just speed) because forces have directions, and the direction of motion determines what forces do to the body (speed up, slow down, change direction). Speed alone cannot distinguish these.**

Distinguish scalar speed (magnitude only) from vector velocity (magnitude + direction). Explain why direction is physically essential.

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Q4. A policeman moving on a highway with a speed of 25 km/h fires a bullet at a thief's car speeding away in the same direction at 100 km/h. With what speed does the bullet hit the thief's car if the muzzle speed of the bullet is 120 km/h?

**Answer: Velocity of bullet relative to the ground = velocity of bullet relative to policeman's car + velocity of policeman's car relative to ground =  $120 + 25 = 145$  km/h. Velocity of bullet relative to thief's car = velocity of bullet relative to ground – velocity of thief's car relative to ground =  $145 - 100 = 45$  km/h. The bullet hits the thief's car at 45 km/h.**

Two-step relative-velocity calculation: (i) bullet velocity in ground frame, (ii) bullet velocity in thief's-car frame.

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Q5. Measure the height of the roof of your house from the ground. Drop a stone from the roof and let it fall freely. Calculate (i) the distance travelled in 3 s, (ii) the velocity of the stone when it reaches the ground, (iii) the velocity at 2 s after the start. (Take  $g = 10 \text{ m/s}^2$ .)

**Answer: Assume roof height  $h = 20 \text{ m}$  (a typical one-storey house; substitute your measured value). (i) Distance in 3 s, starting from rest:  $s = \frac{1}{2} g t^2 = \frac{1}{2}(10)(9) = 45 \text{ m}$ . (NB: if your house is less than 45 m tall, the stone hits ground BEFORE 3 s — in that case  $t_{\text{ground}} = \sqrt{2h/g} < 3 \text{ s}$ .) (ii) Velocity at ground:  $v^2 = 2 g h = 2 \times 10 \times 20 = 400$ , so  $v = 20 \text{ m/s}$  downward. (iii) Velocity at 2 s:  $v = g t = 10 \times 2 = 20 \text{ m/s}$  downward (still in the air assuming  $h > 20 \text{ m}$ ).**

Apply the equations of free fall starting from rest. Distance =  $\frac{1}{2}gt^2$ ; velocity =  $gt$ ; for impact use  $v^2 = 2gh$ .

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Q6. Drop a ball with full force on the ground and note down its displacement with time. (a) Plot displacement vs time graph. (b) Plot qualitatively velocity vs time graph. (c) Plot qualitatively acceleration vs time graph.

**Answer:** (a) Displacement vs time graph: the ball falls, hits the ground, bounces back up to a smaller height, falls again, bounces to even smaller height, and so on. The x-t graph looks like a series of parabolic arcs of decreasing peak height, with cusps at each bounce (instant of contact with the ground). (b) Velocity vs time graph: between bounces, velocity changes linearly (slope =  $-g$  during descent,  $+g$  after the bounce reverses direction). At each bounce, velocity jumps suddenly from negative (downward) to positive (upward), with the magnitude reducing each time. The graph is a sequence of straight-line segments with discontinuous jumps at the bounce times. (c) Acceleration vs time graph: between bounces, acceleration is constant at  $-g$  (gravity). At each bounce, there is a brief but very large positive acceleration (from the ground's normal force during the very short contact time). So the graph is mostly a horizontal line at  $-g$ , with narrow upward spikes at each bounce.

Bounce introduces a near-instantaneous direction reversal. The key insight: between bounces, the motion is free fall; the bounces appear as discontinuities in velocity and as impulsive spikes in acceleration.

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Q7. Observe a rain cloud. Suppose the cloud is at an altitude of 1 km above the ground. (a) If a raindrop falls from such a height freely under gravity, what will be the speed of the rain in SI units? (b) Convert the calculated speed to km/h. (Take  $g = 10 \text{ m/s}^2$ .)

**Answer:** (a) Free fall from rest from height  $h = 1000 \text{ m}$ :  $v^2 = 2gh = 2 \times 10 \times 1000 = 20000$ , so  $v = \sqrt{20000} \approx 141.4 \text{ m/s}$ . (b)  $v$  in km/h =  $141.4 \times 3600 / 1000 = 509.1 \text{ km/h}$ . Interpretation: This is much faster than the actual speed of falling rain. Real raindrops reach a terminal velocity of only about 10 m/s because of air resistance. The free-fall calculation overestimates the speed by a factor of  $\sim 14$  — demonstrating why neglecting air resistance is a poor approximation for small falling objects.

Apply  $v^2 = 2gh$ . Then convert m/s to km/h by multiplying by 3.6. Note the comparison with the real terminal velocity — air resistance matters.

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Q8. If you and a friend ride a bicycle on the road, you will notice that at different times the two bicycles are at different positions. Note down five positions at five time intervals for each. (a) Plot the position-time graph for uniform motion. (b) Plot the position-time graph for non-uniform motion. (c) Calculate velocity from a position-time graph. (d) Calculate the distance travelled by you and your friend in 10 minutes. (e) Examine whose speed is more.

**Answer:** Sample data — you ride at 12 km/h uniformly; your friend accelerates from rest. Your positions at  $t = 0, 2, 4, 6, 8, 10 \text{ min}$ : 0, 0.4, 0.8, 1.2, 1.6, 2.0 km. Friend's positions at the same times (accelerated, starts at rest, reaches 24 km/h at 10 min): 0, 0.04, 0.16, 0.36, 0.64, 1.0 km (very approximate). (a) Uniform motion graph: a straight line with constant slope (= velocity). For you: line from (0, 0) to (10, 2) km. (b) Non-uniform motion graph: a curve with increasing slope — for friend's accelerated motion, an upward parabola. (c) Velocity = slope of x-t graph =  $\Delta x / \Delta t$ . For you:  $2 \text{ km} / 10 \text{ min} = 12 \text{ km/h}$ . (d) You: 2 km in 10 min. Friend: 1 km in 10 min (varies). (e) Average speed of you (12 km/h) > friend's average speed (6 km/h). Friend's INSTANTANEOUS speed at 10 min (24 km/h) is greater than yours — depending on which is meant by 'speed', the answer differs.

Look for correct graph shapes (line for uniform, curve for non-uniform), velocity-from-slope, and a thoughtful comparison of average vs instantaneous speed.

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